

Continuous Distributions

Unlike in the discrete case, **continuous distribution** can assume any value in an interval on the real line (bounded or unbounded).

"support" of the dist $\rightarrow$ range of possible values

To describe a continuous distribution of random variable X , we often use the **probability density function (pdf)**, or just "density," denoted f_X .

The density f_X has the following properties:

1. $f_X(x) \geq 0$ for all x

2.

$$\int_{-\infty}^{\infty} f_X(u) du = 1$$

3. For any $a \leq b$,

$$P(a \leq X \leq b) = \int_a^b f_X(u) du$$

4. The expected value of X is calculated as

$$E(X) = \int_{-\infty}^{\infty} u f_X(u) du$$

Comments

1. The set of values x for which $f_X(x) > 0$ is called the **support** for the distribution of the random variable X .

2. Note that Condition 3 above implies that $P(X = a) = 0$ for all $a \in \mathbb{R}$.
Hence,

$$P(a < X \leq b) = P(a \leq X < b) = P(a < X < b) = P(a \leq X \leq b).$$

3. The CDF is defined as above, and all of the properties given above hold. Note, however, that since $P(X = a) = 0$ for all a , then $F_X(x)$ must be continuous. Also, from the Fundamental Theorem of Calculus,

$$f_X(x) = \frac{\partial}{\partial x} F_X(x).$$

4. In order for $E(X)$ to exist², need either

$$\int_0^{\infty} u f_X(u) du < \infty$$

or

$$\int_{-\infty}^0 u f_X(u) du > -\infty.$$

Otherwise, one is faced with defining $\infty - \infty$. The Cauchy Distribution is a classic example for which $E(X)$ does not exist. In that case,



$$f_X(x) = \frac{1}{\pi(1+x^2)}, \quad -\infty < x < \infty.$$

$$E(X) = \int_{-\infty}^{\infty} u \frac{1}{\pi(1+u^2)} du$$

$$\int_0^{\infty} \frac{1}{u} du = \infty$$

²We stress that " $E(X)$ exists" is not the same as " $E(X)$ is finite." If $E(X) = \infty$, then $E(X)$ does exist.

1 Rule for the Lazy Statistician

2 The Rule for the Lazy Statistician in the continuous case is

$$3 \quad E(g(X)) = \int_{-\infty}^{\infty} g(u) f_X(u) du.$$

4 Again, it is worth stressing that this is **not** the definition of $E(g(X))$. It is
5 instead a **theorem** that builds on some sophisticated probability theory.

6 To **define** $E(g(X))$, one could start by letting $Y = g(X)$ and then derive the
7 pdf for Y , call it f_Y , and then

$$Y(\omega) = g(X(\omega))$$

$$8 \quad E(g(X)) = E(Y) = \int_{-\infty}^{\infty} u f_Y(u) du.$$

9 Hence, one is “lazy” in that it is possible to find $E(g(X))$ without deriving
10 f_Y . **But**, in order to calculate probabilities of the form

$$11 \quad P(a \leq g(X) \leq b) = P(a \leq Y \leq b)$$

12 one would need to derive f_Y .

Exercise: The distribution function (CDF) for a continuous random variable X is given by

$$F_X(x) = \begin{cases} 1 - e^{-kx^2} & \text{for } x \geq 0 \\ 0 & \text{for } x < 0 \end{cases}$$

~~$$F_X(x) = 1 - e^{-kx^2}$$~~

where k is a positive constant.

1. Determine the density function $f_X(x)$.
2. Show that regardless of the value of $k > 0$ this is a legitimate density function.
3. For $k = 1$, find $P(2 < X < 4)$ and $P(X > 2)$.
4. Find $E(X)$ for any value of k .
5. Find $E(1/X)$ for any value of k .

$$\textcircled{1} f_X(x) = \frac{d}{dx} F_X(x) = \begin{cases} 2xk e^{-kx^2}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

$$\textcircled{2} \text{ Verify that } f_X(x) \geq 0 \text{ for all } x \text{ and that } \int_{-\infty}^{\infty} f_X(u) du = 1$$

$$\textcircled{3} k=1$$

$$P(2 < X < 4) = \int_2^4 f_X(u) du = \int_2^4 2u e^{-ku^2} du = 0.018$$

Or:

$$P(2 < X \leq 4) = F_X(4) - F_X(2)$$

$$= P(2 < X \leq 4)$$

$$\textcircled{4} \quad E(X) = \int_{-\infty}^{\infty} u f_X(u) du$$

$$= \int_{-\infty}^{\infty} u 2u k e^{-ku^2} du$$

$$= 2k \int_0^{\infty} u^2 e^{-ku^2} du$$

use integral table, 2nd row, 3rd col.

$$= 2k \left(\frac{\Gamma\left(\frac{3}{2}\right)}{2k^{3/2}} \right) = \frac{\sqrt{\pi}}{2\sqrt{k}}$$

$$\Gamma\left(\frac{3}{2}\right) = \frac{1}{2} \Gamma\left(\frac{1}{2}\right) = \frac{\sqrt{\pi}}{2}$$

$$(5) \quad E\left(\frac{1}{x}\right) \neq \frac{1}{E(x)}$$

$$E(X) = \int$$

$$E\left(\frac{1}{x}\right) = \int_{-\infty}^{\infty} \frac{1}{u} f_X(u) du$$

Rule for
Lazy Stat.

$$= \int_0^{\infty} 2k e^{-ku^2} du$$

$$= 2k \int_0^{\infty} e^{-ku^2} du$$

$$= 2k \left(\frac{1}{2} \frac{\sqrt{\pi}}{\sqrt{k}} \right) \quad \text{Using integral table}$$

$$= \sqrt{\pi k}$$

$$Y = \frac{1}{X}$$

$$F_Y(y) = P(Y \leq y) = P\left(\frac{1}{X} \leq y\right)$$

$$= P\left(X \geq \frac{1}{y}\right) = 1 - F_X\left(\frac{1}{y}\right)$$

$$= e^{-k/y^2}, \text{ for } y \geq 0$$

$$(\text{For } y \leq 0, F_Y(y) = 0)$$

$$f_Y(y) = \frac{\partial}{\partial y} e^{-k/y^2}$$

Or:

$$E(Y) = \int_0^{\infty} P(Y > u) du$$

(see exercises
for result)

Exercise: The density function for a continuous random variable X is given by:

$$f_X(x) = \begin{cases} 4x & 0 \leq x < \frac{1}{2} \\ 4 - 4x & \frac{1}{2} \leq x < 1 \\ 0 & \text{otherwise} \end{cases}$$

1. Determine a formula for the cdf of X , $F_X(x)$.

2. Determine $P(0.25 < X \leq 0.75)$

3. Find $E(X)$.

“Named” Continuous Distributions

There are a wide range of continuous distributions that are important in Statistics. A few important ones will be presented now, but others will appear later, when needed. The ones shown here:

1. The Uniform(a, b) Distribution
2. The Pareto(α, λ) Distribution
3. The Exponential(λ) Distribution
4. The Gamma(α, β) Distribution
5. The Normal(μ, σ^2) Distribution

1 The Uniform(a, b) Distribution

2 If one says that X has the **Uniform(a, b) distribution**, this means that

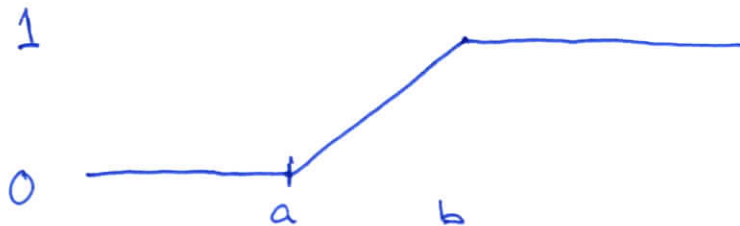
$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a < x < b \\ 0, & \text{otherwise} \end{cases}$$


4 Heuristically, any value between a and b is “equally likely.” (Of course,
5 this is not a technically correct way of thinking about it, as all values have
6 probability zero under a continuous distribution.)

7 It is not difficult to show that $E(X) = (a + b)/2$ and $V(X) = (b - a)^2/12$.

8 Further,

$$F_X(x) = \begin{cases} 0, & x \leq a \\ \frac{x-a}{b-a}, & a < x < b \\ 1, & x \geq b \end{cases}$$



- 1 **Exercise:** Take a stick of unit length and break it at random. What is the
 2 distribution of the length of the longer piece?



$$U \sim \text{Uniform}(0, 1)$$

L = length of longer piece

Derive the CDF for L . Know $0.5 \leq L \leq 1$.

$$F_L(l) = 0 \text{ if } l < 0.5$$

$$F_L(l) = 1 \text{ if } l > 1$$

For $0.5 \leq l \leq 1$, when is $L \leq l$? Or, when $L > l$?



$$L \leq l \text{ when } 1-l \leq U \leq l$$

$$F_L(l) = P(L \leq l) = P(1-l \leq U \leq l)$$

$$= \int_{1-l}^l 1 \, du = l - (1-l) = 2l - 1$$

$$F_X^{-1}(U) \sim X$$

Conclude the CDF is

$$F_L(l) = \begin{cases} 0, & l < 0.5 \\ 2l-1, & 0.5 \leq l \leq 1 \\ 1, & l > 1 \end{cases}$$

matches CDF of Uniform(0.5, 1) dist.

By uniqueness of CDF, $L \sim \text{Uniform}(0.5, 1)$

- 1 **Exercise (Based on HOTS):** Consider the "broken-stick-into-a-triangle"
 2 question. What is the distribution of the length of the longest of the three
 3 pieces (regardless of whether or not a triangle can be formed)?

4 $U_1, U_2 \sim \text{Uniform}(0,1)$ positions of cuts
 5 assume U_1, U_2 independent

6 $L = \text{length of longest piece}$. What is F_L ?

7 know $\frac{1}{3} \leq L \leq 1$, so $F_L(l) = \begin{cases} 0, & l < \frac{1}{3} \\ ? , & \frac{1}{3} \leq l \leq 1 \\ 1, & l > 1 \end{cases}$

10 Fix $\frac{1}{3} \leq l \leq 1$. When is $L \leq l$? When is $L > l$?

12 (1)



14 both $U_1, U_2 > l$

16 (2)

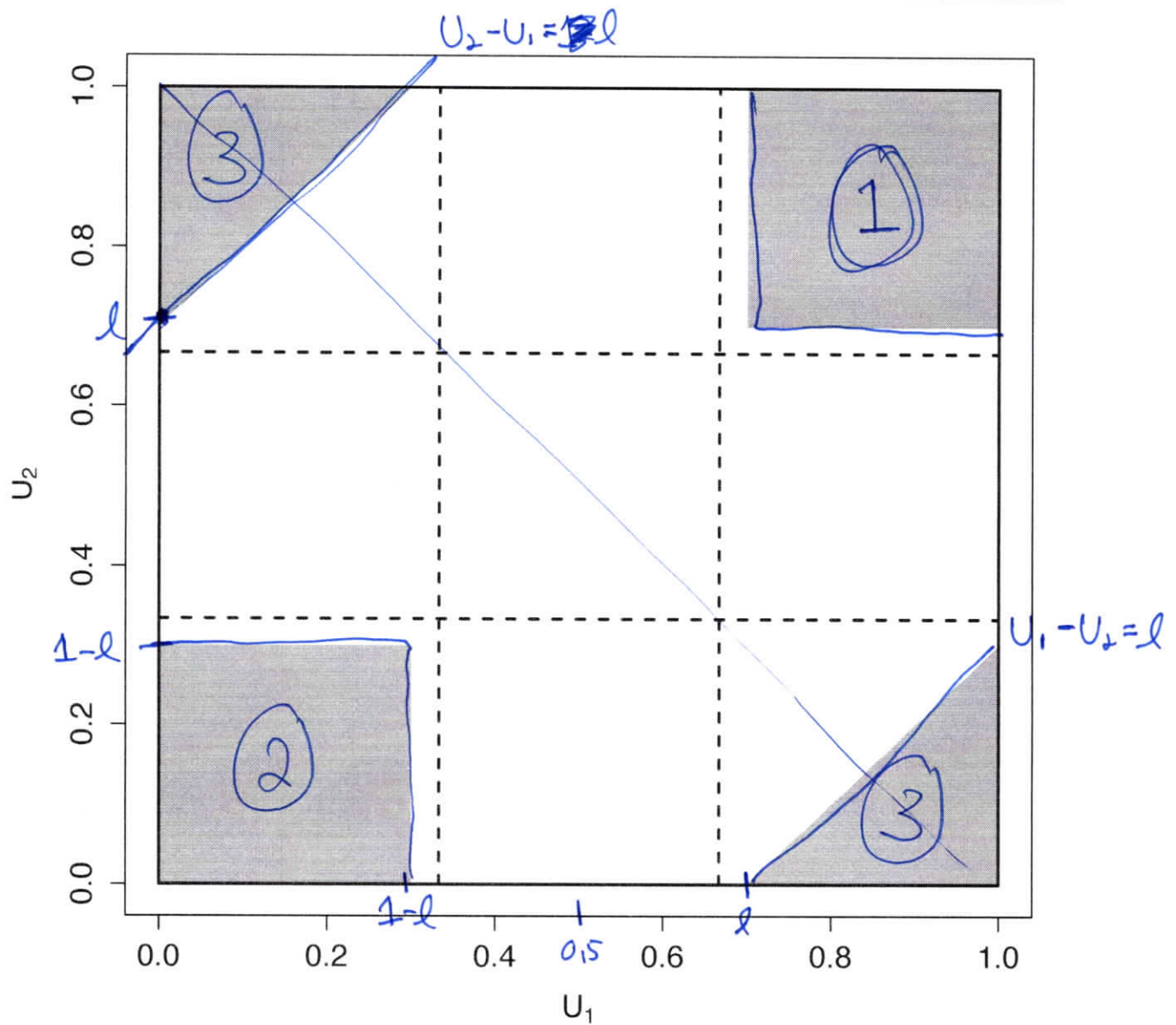


18 both $U_1, U_2 < 1-l$

20 (3)



22 $|U_1 - U_2| > l$

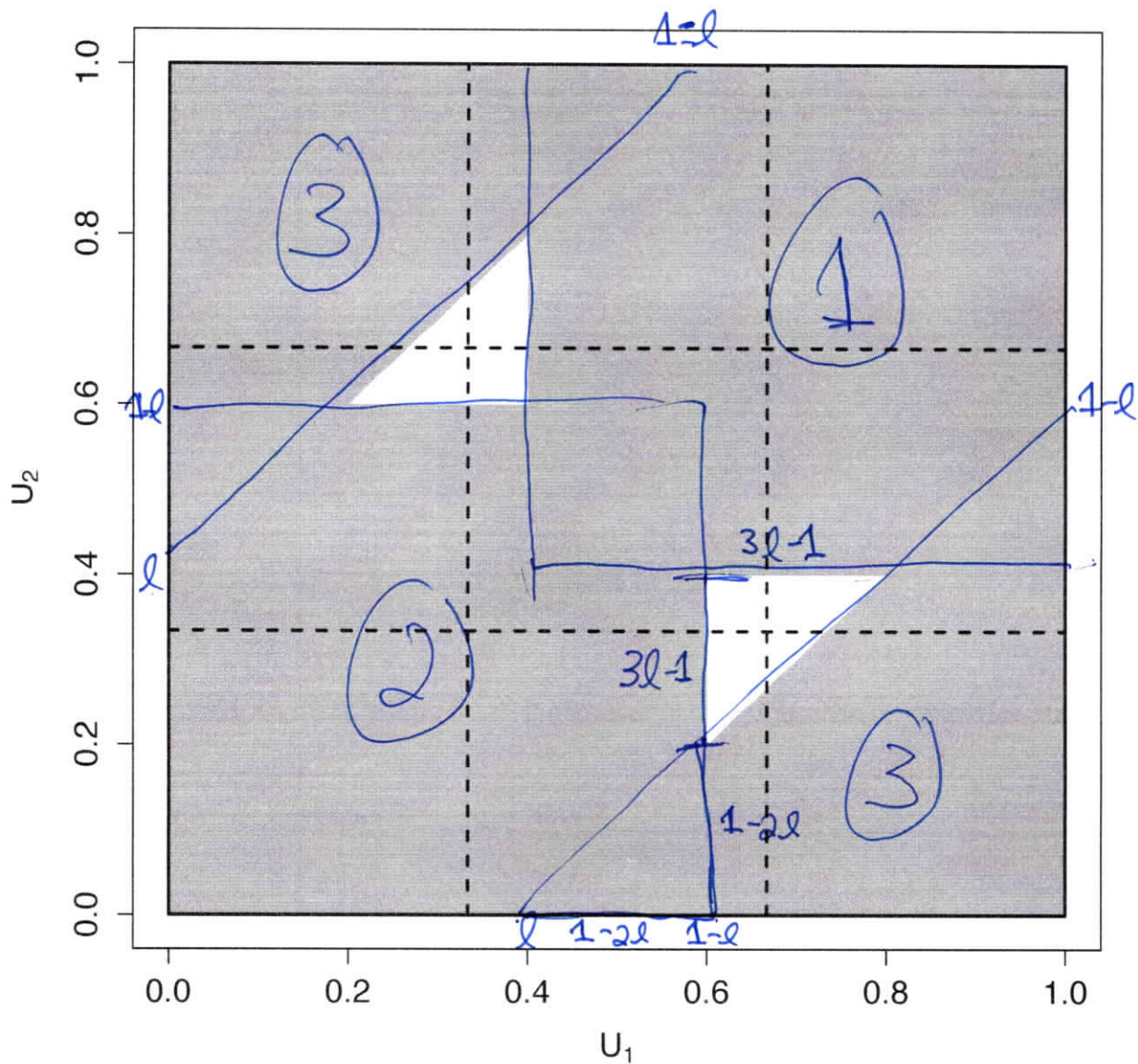


Case where $\frac{1}{2} \leq l \leq 1$

$P(L > l) = \text{area of shaded region}$

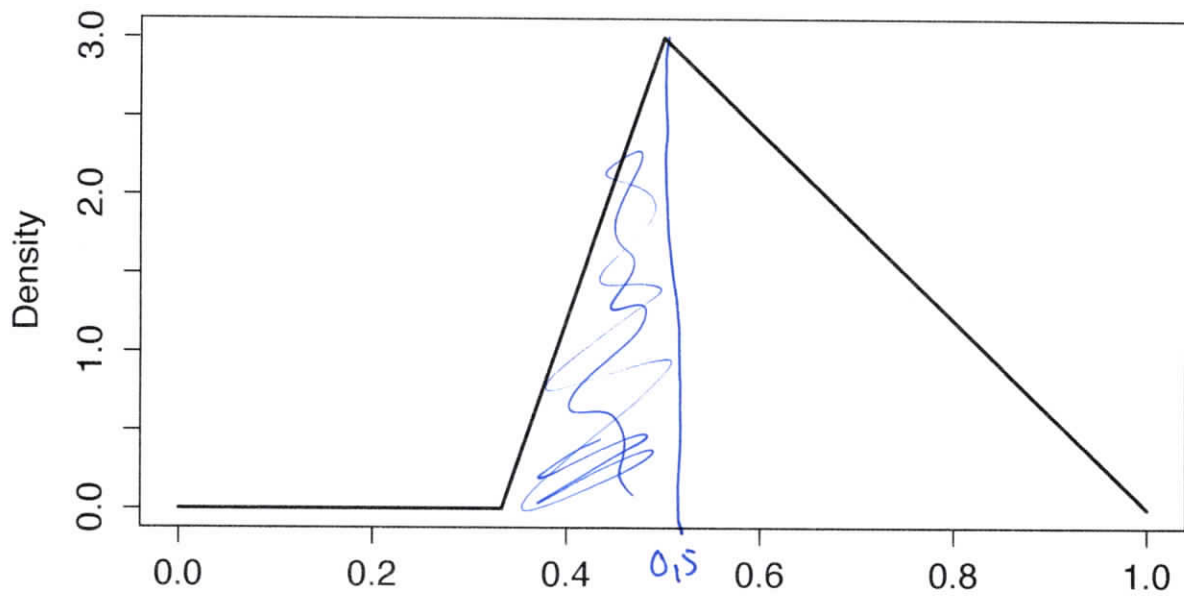
$$= 3(1-l)^2$$

$$P(L \leq l) = 1 - 3(1-l)^2$$



Case where $\frac{1}{3} \leq l \leq \frac{1}{2}$

$$P(L \leq l) = \text{area of unshaded region} \\ = (3l - 1)^2$$



$$F_L(l) = \begin{cases} 0, & l < \frac{1}{3} \\ (3l-1)^2, & \frac{1}{3} \leq l < \frac{1}{2} \\ 1 - 3(1-l)^2, & \frac{1}{2} \leq l \leq 1 \\ 1, & l > 1 \end{cases}$$

$$f_L(l) = \begin{cases} 6(3l-1), & \frac{1}{3} \leq l < \frac{1}{2} \\ 6(1-l), & \frac{1}{2} \leq l \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

$$P(\text{form a } \triangle) = F_L\left(\frac{1}{2}\right) = \frac{1}{4}$$

1 The Pareto(α, λ) Distribution

2 The random variable X has the **Pareto(α, λ) distribution** if $\alpha > 0$ and $\lambda > 0$
3 with

$$4 \quad f_X(x) = \begin{cases} \frac{\alpha \lambda^\alpha}{x^{\alpha+1}}, & x \geq \lambda \\ 0, & \text{otherwise} \end{cases}.$$

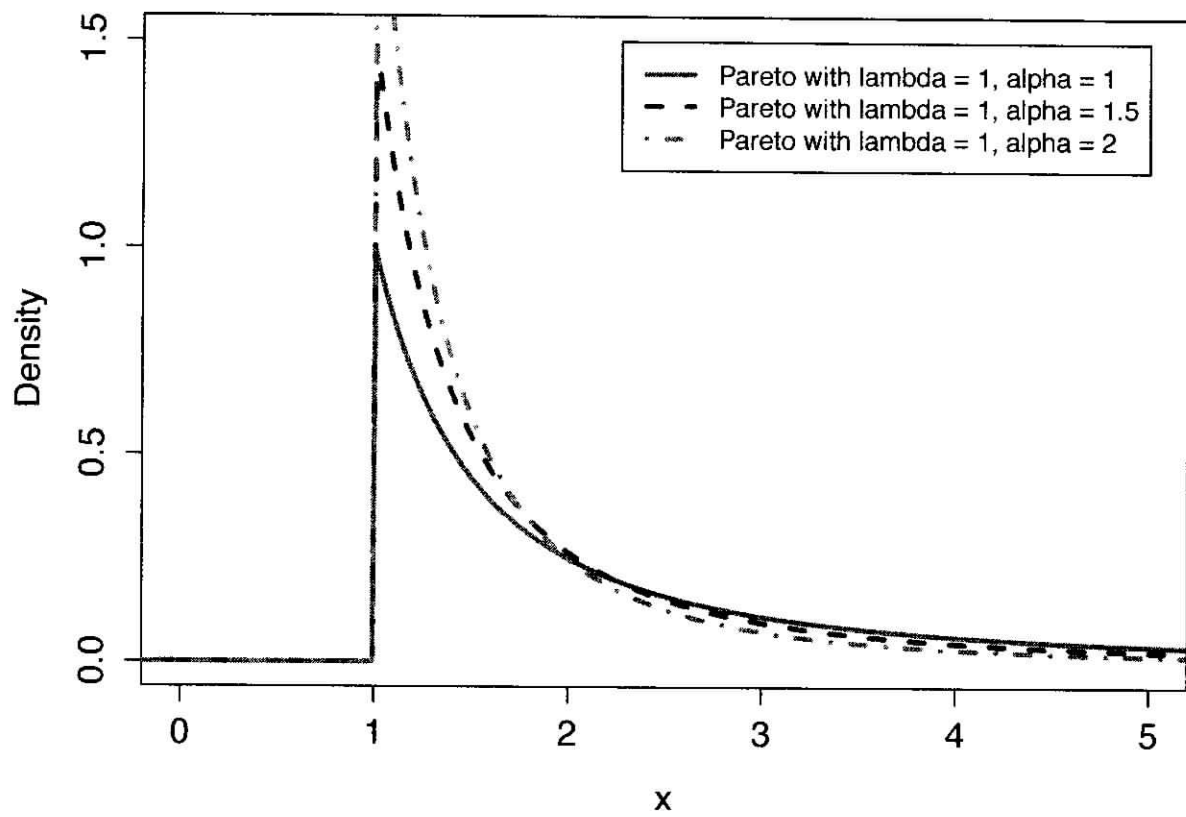
5 This is often used for modelling the tail of a distribution.

6 Some useful properties:

$$7 \quad E(X) = \begin{cases} \infty, & \alpha \leq 1 \\ \frac{\alpha \lambda}{\alpha - 1}, & \alpha > 1 \end{cases}$$

$$8 \quad V(X) = \begin{cases} \text{undefined}, & \alpha \leq 1 \\ \infty, & 1 < \alpha \leq 2 \\ \frac{\alpha \lambda}{(\alpha - 1)^2 (\alpha - 2)}, & \alpha > 2 \end{cases}$$

$$9 \quad F_X(x) = \begin{cases} 1 - \left(\frac{\lambda}{x}\right)^\alpha, & x \geq \lambda \\ 0, & x < \lambda \end{cases}$$



1 The Exponential(λ) Distribution

2 When X has the **Exponential(λ) distribution**, the following hold:

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

"rate parameterization"

$$E(X) = 1/\lambda \text{ and } V(X) = 1/\lambda^2$$

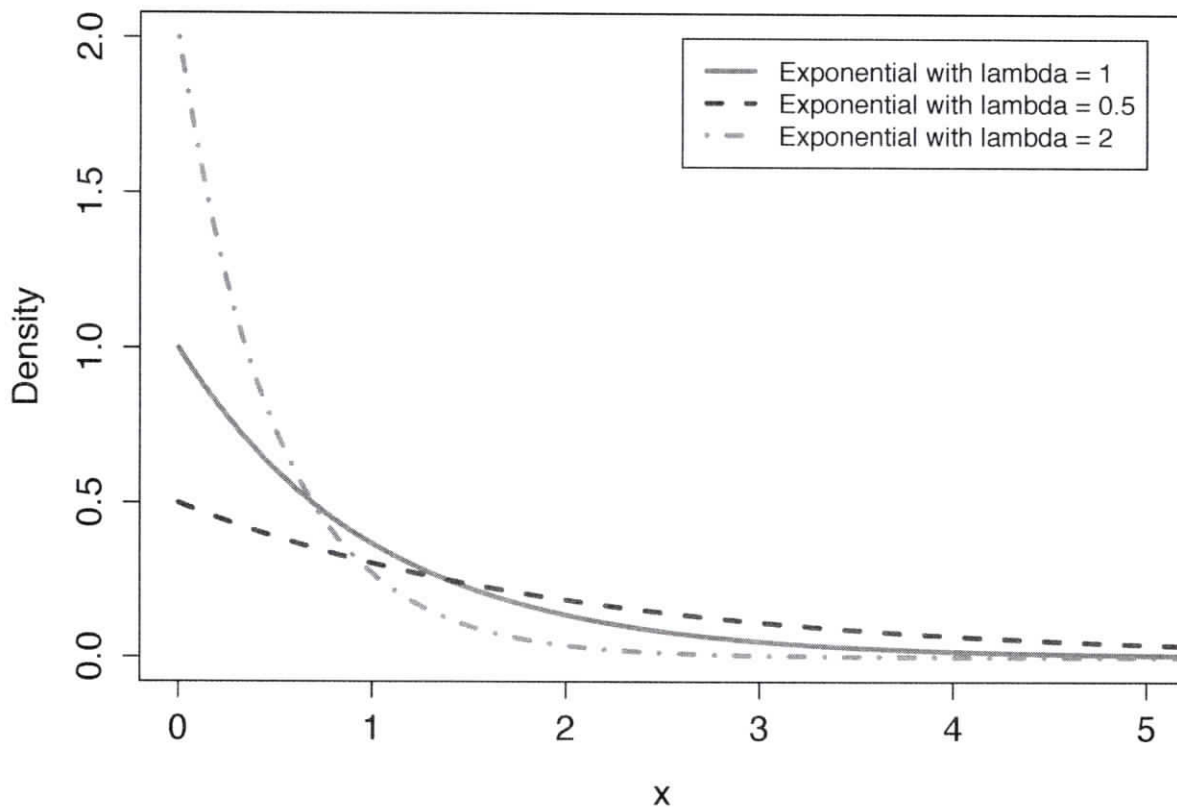
4

$$f_X(x) = \frac{1}{\beta} e^{-x/\beta}$$

"mean parameterization"

$$E(X) = \beta$$

$$F_X(x) = \begin{cases} 1 - e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

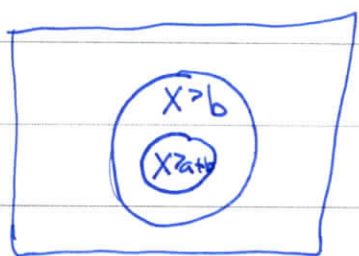


Exercise: Suppose that X has the Exponential(λ) distribution, and that $a > 0$ and $b > 0$. Show that

$$P(X > a + b | X > b) = P(X > a).$$

Why is this called the "Memoryless Property?"

$$P(X > a + b | X > b) = \frac{P(X > a + b \text{ and } X > b)}{P(X > b)}$$



$$= \frac{P(X > a + b)}{P(X > b)}$$

$$= \frac{1 - F_X(a + b)}{1 - F_X(b)}$$

$$= \frac{e^{-\lambda(a+b)}}{e^{-\lambda b}}$$

$$= e^{-\lambda a}$$

$$= P(X > a)$$

e.g. If I model the times between "events" (jumps) as exponential, the prob. the next event is at least a time units away does not depend on time since previous event "memoryless property"

1 The Gamma(α, β) Distribution

2 For $\alpha > 0$ and $\beta > 0$, random variable X has the **Gamma(α, β) distribu-**
3 **tion** when

$$f_X(x) = \begin{cases} \frac{x^{\alpha-1} e^{-\beta x} \beta^\alpha}{\Gamma(\alpha)}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

5 where

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx \text{ is the } \mathbf{\text{Gamma Function}}.$$

7 **Properties worth noting:** If $X \sim \text{Ga}(\alpha, \beta)$, then

8 1. $E(X) = \alpha/\beta$

9 2. $V(X) = \alpha/\beta^2$

10 3. If $\alpha = 1$, then $X \sim \text{Exp}(\beta)$.

11 4. For any $c > 0$, the random variable $cX \sim \text{Ga}(\alpha, \beta/c)$.

12 5. If $X_1, X_2, \dots, X_n$ are independent random variables with $X_i \sim \text{Ga}(\alpha_i, \beta)$,
13 then $\sum_i X_i \sim \text{Ga}(\sum_i \alpha_i, \beta)$.

14 6. The sum of n independent Exponential(β) random variables has the
15 Gamma(n, β) distribution. This is called the **Erlang Distribution**.

